

ME315_Lab04_Elisa

Lab 04

Verificando diretórios e os caminhos

```
getwd()
```

```
[1] "C:/Users/elisa/Downloads/ME315_Lab04"
```

```
list.files()
```

```
[1] "data"                                "ME315_Lab04_Elisa.qmd"
[3] "ME315_Lab04_Elisa.rmarkdown"
```

1.

```
library(readxl)
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

```
✓ dplyr      1.2.1    ✓ readr      2.1.5
✓ forcats    1.0.1    ✓ stringr    1.5.1
✓ ggplot2     3.5.1    ✓ tibble     3.2.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr       1.2.2
```

— Conflicts — tidyverse_conflicts() —

```
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()     masks stats::lag()
```

ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

2.

```
fname <- "data/WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

3.

```
countries <- read_excel(fname, sheet = "Country")
```

4.

```
countries %>% select(`Short Name`) %>% print(n = 10)
```

```
# A tibble: 263 × 1
  `Short Name`
  <chr>
```

```

1 Aruba
2 Afghanistan
3 Angola
4 Albania
5 Andorra
6 Arab World
7 United Arab Emirates
8 Argentina
9 Armenia
10 American Samoa
# i 253 more rows

```

Fazendo o filtro

```

countries <- countries %>%
  filter(`Short Name` %in% c("Brazil", "United States", "Argentina", "China", "India"))

```

5.

```

WDI <- read_excel(fname, sheet = "Data")

```

6.

```

WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, as.character(1960:2018))

```

```

glimpse(WDI)

```

Rows: 422,136

Columns: 61

```

$ `Country Code`      <chr> "ARB", "ARB", "ARB", "ARB", "ARB", "ARB", "ARB", "ARB...
$ `Indicator Code`    <chr> "PA.NUS.PPP.05", "PA.NUS.PRVT.PP.05", "EG.CFT.ACCS.ZS...
$ `1960`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1961`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1962`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1963`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1964`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1965`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1966`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1967`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1968`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1969`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1970`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1971`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1972`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1973`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1974`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1975`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1976`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1977`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1978`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1979`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...

```

```

$ `1980` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1981` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1982` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1983` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1984` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1985` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1986` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1987` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1988` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1989` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1990` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1991` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1992` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1993` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1994` <dbl> NA, NA, NA, NA, 5.455825e+01, NA, NA, NA, NA, NA, NA,...
$ `1995` <dbl> NA, NA, NA, NA, 5.494440e+01, NA, NA, NA, NA, NA, NA,...
$ `1996` <dbl> NA, NA, NA, NA, 5.693231e+01, NA, NA, NA, NA, NA, NA,...
$ `1997` <dbl> NA, NA, NA, NA, 5.802529e+01, NA, NA, NA, NA, NA, NA,...
$ `1998` <dbl> NA, NA, NA, NA, 5.972716e+01, NA, NA, NA, NA, NA, NA,...
$ `1999` <dbl> NA, NA, NA, NA, 6.096064e+01, NA, NA, NA, NA, NA, NA,...
$ `2000` <dbl> NA, NA, 7.370250e+01, NA, 5.966136e+01, NA, NA, NA, N...
$ `2001` <dbl> NA, NA, 7.494486e+01, NA, 6.294551e+01, NA, NA, NA, N...
$ `2002` <dbl> NA, NA, 7.625807e+01, 7.526257e+01, 6.094426e+01, NA,...
$ `2003` <dbl> NA, NA, 7.712046e+01, 7.665350e+01, 6.322675e+01, NA,...
$ `2004` <dbl> NA, NA, 7.809444e+01, 7.902110e+01, 6.409467e+01, 9.3...
$ `2005` <dbl> NA, NA, 7.906200e+01, 8.020928e+01, 6.641386e+01, 9.3...
$ `2006` <dbl> NA, NA, 7.985168e+01, 8.411696e+01, 7.181785e+01, 9.4...
$ `2007` <dbl> NA, NA, 8.057054e+01, 8.286337e+01, 6.906648e+01, 9.4...
$ `2008` <dbl> NA, NA, 8.121089e+01, 8.508341e+01, 7.220289e+01, 9.5...
$ `2009` <dbl> NA, NA, 8.180963e+01, 8.467446e+01, 7.181828e+01, 9.4...
$ `2010` <dbl> NA, NA, 8.236810e+01, 8.600762e+01, 7.346665e+01, 9.5...
$ `2011` <dbl> NA, NA, 8.278329e+01, 8.642827e+01, 7.394210e+01, 9.5...
$ `2012` <dbl> NA, NA, 8.312030e+01, 8.707058e+01, 7.524410e+01, 9.5...
$ `2013` <dbl> NA, NA, 8.353346e+01, 8.817684e+01, 7.716230e+01, 9.6...
$ `2014` <dbl> NA, NA, 8.389760e+01, 8.734274e+01, 7.553898e+01, 9.5...
$ `2015` <dbl> NA, NA, 8.417160e+01, 8.913012e+01, 7.874115e+01, 9.6...
$ `2016` <dbl> NA, NA, 8.451017e+01, 8.967869e+01, 7.966564e+01, 9.6...
$ `2017` <dbl> NA, NA, NA, 9.027369e+01, 8.074929e+01, 9.700397e+01,...
$ `2018` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...

```

7. Por questão de memória, vamos filtrar os países antes do pivot longer

```

WDI <- WDI %>%
  filter(`Country Code` %in% countries$`Country Code`)

```

```

WDI <- WDI %>%
  pivot_longer(
    cols      = as.character(1960:2018),
    names_to  = "YEAR",
    values_to = "value"
  )

glimpse(WDI)

```

Rows: 471,705

Columns: 4

```
$ `Country Code` <chr> "ARG", "ARG", "ARG", "ARG", "ARG", "ARG", "ARG", "ARG..."
$ `Indicator Code` <chr> "PA.NUS.PPP.05", "PA.NUS.PPP.05", "PA.NUS.PPP.05", "P..."
$ YEAR <chr> "1960", "1961", "1962", "1963", "1964", "1965", "1966..."
$ value <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
```

8.

```
class(WDI$YEAR)
```

[1] "character"

```
WDI <- WDI %>%
  mutate(YEAR = as.integer(YEAR))

class(WDI$YEAR)
```

[1] "integer"

9.

```
WDI <- WDI %>%
  pivot_wider(
    names_from = `Indicator Code`,
    values_from = value
  )

WDI <- WDI %>%
  left_join(countries %>% select(`Country Code`, `Short Name`), by = "Country Code") %>%
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)
```

10.

```
WDI <- WDI %>%
  arrange(`Country Code`, YEAR)
```

11.

```
ggplot(WDI, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line() +
  labs(
    x = "Ano",
    y = "Taxa de natalidade (SP.DYN.CBRT.IN)",
    color = "País"
  ) +
  theme_minimal()
```

Warning: Removed 5 rows containing missing values or values outside the scale range (`geom\_line()`).

